

Aman Sahu

Data Analyst

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SUMMARY

Results-driven Data Analyst with 4+ years of experience in financial, healthcare, and supply chain domains, skilled in SQL, Python, R, Tableau, Power BI, and advanced analytics. Expertise in ETL, data warehousing, predictive modeling, forecasting, and KPI dashboards to drive revenue growth, cost optimization, and efficiency. Adept at analyzing complex datasets, risk assessment, claims analytics, and inventory optimization while ensuring compliance and data governance. Proven collaborator in delivering scalable analytics solutions, achieving 20% process improvement, 15% error reduction, and enhanced decision-making.

TECHNICAL SKILLS

Databases & Management:	SQL, NoSQL (MongoDB, Cassandra), ETL Processes
Methodologies:	SDLC, Agile, Waterfall, A/B Testing, Data Governance, Data Security Practices (Encryption, Masking), Data Privacy Laws (GDPR, HIPAA)
Programming Languages:	Python, Scala, R, SQL, SAS
Data Processing & ETL Tools:	Talend, Informatica, SSIS, Apache Airflow
Data Warehousing:	Amazon Redshift, Snowflake, Google BigQuery
Cloud Platforms:	AWS, Azure
Data Visualization:	Tableau, Power BI, Excel (Advanced, VBA), Crystal Reports, SSRS
Statistical Analysis & Machine Learning:	SAS, SPSS, TensorFlow, Scikit-learn, Data Mining, ANOVA, Clustering, Classification, Regression, Forecasting & Modeling, Predictive Modeling
Business Analyst Tools & Techniques	Microsoft Excel (Pivot Tables, Power Query, VBA), Power BI, Tableau, SQL, SAS, SPSS, Google Analytics, Lucidchart, Microsoft Visio, JIRA, Confluence, KPI Dashboards, Data Modeling, Requirement Gathering, Root Cause Analysis, Reporting & Documentation,
Version Control:	Git/GitHub
Other Technical Skills:	Jira, Confluence, Data Quality Tools, Data Cleaning, Data Wrangling, Process Mapping, Solution-Oriented Analysis, Ad Hoc Analysis, Project Management, , Root Cause Analysis.
Certification	<u>Certified Scrum Product Owner</u>

EDUCATION

Master of Science in Information Technology

Worcester Polytechnic Institute (Worcester, Massachusetts, USA).

Bachelor of Technology in Computer Science

Rajiv Gandhi Proudyogiki Vishwavidyalaya, Bhopal, MP India.

WORK EXPERIENCE

Data Analyst | Optum, TX

Aug 2024 – Present

- Extracted and analyzed EHR and operational data using **SQL and Python** to evaluate bed occupancy and staff utilization, pinpointing capacity bottlenecks that slashed average patient wait times by 22% across 5 units.
- Developed **time-series forecasting** models in R to predict bed occupancy and staff demand peaks, enabling proactive scheduling that cut overtime expenses by 18% across 8 units.
- Developed **Power BI reports** with DAX measures for drill-down analysis of capacity metrics, enabling stakeholders to track performance variances and achieve 17% faster decision cycles.
- Applied regression analysis and **ANOVA tests** in R to correlate ER throughput with staffing ratios, identifying inefficiencies that improved patient flow by 20% during high-volume periods.
- Recommended process improvements based on **Alteryx-driven simulations** and Snowflake forecasting models of staffing scenarios, optimizing shift allocations and minimizing overcrowding by 25%.
- Conducted **cohort analysis** and **data mining** on readmission data using Azure Synapse, uncovering patterns that informed targeted interventions reducing 30-day readmissions by 15%.
- Partnered with operations **and clinical staff to define KPIs** and refine data models, translating business needs into analytical solutions that aligned with **HIPAA standards**.
- Performed **hypothesis testing** (t-tests and chi-square) in R on staffing interventions, validating statistical significance of capacity improvements that reduced overcrowding.

Data Analyst | Cognizant, India

Nov 2020 – July 2023

- Executed detailed data audits across transactional datasets using **PostgreSQL queries and Google BigQuery** to uncover revenue gaps from billing discrepancies, return fraud, and unclaimed rebates, reclaiming \$300K in annual revenue
- Automated data quality checks and reconciliation workflows via **Python scripts** integrated with **Amazon Redshift** stored procedures, resolving 42% of inter-system variances in rebate tracking.
- Authored **Functional Requirements Document (FRD)** detailing leakage detection logic and margin calculation workflows, incorporating stakeholder inputs via Jira epics, which served as blueprint
- Performed business process modeling with **Visio flowcharts** embedded in BRD, **simulating VBA Excel** scenarios on cleaned datasets to validate margin recovery paths, resulting in approved process redesigns boosting efficiency by 18%.
- Facilitated **JAD (Joint Application Design)** sessions with finance SMEs using Miro boards to refine FRD for root cause analysis modules, integrating query specs that identified 10% hidden freight leakage opportunities.

Jr. Data Analyst | Cognizant, India

Dec 2019 – Oct 2020

- Engineered machine learning pipelines using Python, **Scikit-learn**, and **TensorFlow** to aggregate historical sales data across channels, achieving 25% improvement in forecast accuracy through pattern recognition for seasonal trends and anomaly detection.
- Performed **exploratory data analysis (EDA)** on multi-source datasets (POS, e-commerce, CRM) with **Python (Pandas, NumPy)** and Excel Advanced Analytics to identify demand patterns, resulting in 15% better alignment.
- Implemented real-time data integration with ERP systems, **AWS cloud services**, and APIs for seamless synchronization of inventory levels, reducing stockouts by 30% and enabling what-if scenario modeling for demand planning.
- Designed **customizable dashboards** with **real-time KPI tracking** (e.g., sales velocity, fill rates) using **Tableau and Apache Spark**, providing actionable insights that boosted omnichannel order fulfillment efficiency by 40%.
- Conducted **A/B testing** and scenario analysis on demand **forecasts incorporating external variables** (market trends, seasonality) via R and Jupyter Notebooks, supporting stakeholder presentations that optimized pricing strategies.
- Supported model validation and **UAT (User Acceptance Testing)** for forecasting algorithms by defining success criteria and KPIs, ensuring 45% accuracy in production deployments and reducing forecast variance for high-volume SKUs by 20%.
- Led probabilistic **forecasting and returns prediction models incorporating** external factors like market trends and promotions, minimizing excess inventory costs by 20% via automated reorder calculations.
- Elicited and prioritized requirements from sales, marketing, and supply chain teams using **Agile methodologies and Jira**, creating user stories and process flows that streamlined omnichannel data ingestion.

PROJECTS

HR Analytics Dashboard – Employee Attrition & Satisfaction | Tableau, Excel

- Built an interactive Tableau dashboard from Excel data of 1,470 employees, visualizing attrition trends, demographics, and satisfaction metrics for HR analytics.
- Performed data cleaning, aggregation, and KPI calculations in Excel to prepare structured datasets, ensuring accurate and consistent reporting in Tableau.
- Identified high-risk attrition groups and recommended retention strategies, enabling leadership to make data-driven workforce decisions and improve engagement.

E-Commerce Data Lake Development | AWS S3, Glue, Python, QuickSight

- Designed and implemented an AWS-based data lake using S3 and Glue to store and process transaction data for an e-commerce platform, enabling centralized access to structured and unstructured data.
- Developed ETL workflows using AWS Glue and Python scripts to automate data ingestion, transformation, and cataloging, reducing manual data processing efforts by 40% and improving data accuracy.
- Built interactive dashboards in AWS QuickSight to visualize sales trends and customer purchasing behavior, enabling stakeholders to make data-driven decisions and improve marketing strategies

BI Solutions for E-commerce | Tableau, Excel, Python, Snowflake

- Engineered Tableau dashboards integrating Excel, Shopify, and social media data to monitor KPIs, customer demographics, and sales channels for data-driven inventory, marketing, and operations strategies.
- Orchestrated ETL workflows to extract and load data from Excel, Shopify, and Google Analytics into a Snowflake star-schema warehouse, ensuring clean, scalable, and reliable reporting.
- Deployed Python predictive models (scikit-learn, pandas) to forecast sales and customer trends with 78% accuracy, strengthening demand planning and inventory management.